

EXERCISE 2A

Solution 1: We have

(i) $x^5 - 2x^3 + x + \sqrt{3}$

Since, it is an expression having only non-negative integral powers of x . Also, the highest power of x is 5.

It is a polynomial of degree 5.

(ii) $y^3 + \sqrt{3}y$ it is an expression with non-negative integral powers of y , also has the highest power of y is 3.

$\therefore$ it is a polynomial of degree 3.

(iii) $t^2 - \frac{2}{5}t + \sqrt{5}$ it is an expression with non – negative integral power of t , having highest power of t is 2.

$\therefore$ it is a polynomial of degree 2.

(iv) $x^{100} - 1$

Since, it is an expression having only non-negative integral powers of x . Also, the highest power of x is 100.

It is a polynomial of degree 100.

(v) $\frac{1}{\sqrt{2}}x^2 - \sqrt{2}x + 2$

Since, it is an expression having only non-negative integral powers of x . Also, the highest power of x is 2.

$\therefore$ It is a polynomial of degree 2.

(vi) $x^{-2} + 2x^{-1} + 3$ is not a polynomial. Since, it has a negative power.

(vii) 1 it is a constant polynomial of degree 0.

(viii) $-\frac{3}{5}$ is a constant polynomial of degree 0.

(ix) $\frac{x^2}{2} - \frac{2}{x^2} = \frac{1}{2}x^2 - 2x^{-2}$ it is not a polynomial. Since, it has a negative power.

(x) $\sqrt[3]{2}x^2 - 8$

It is an expression with non-negative integral power of x , having highest power of x is 2.

$\therefore$ it is a polynomial of degree 2.

(xi) $\frac{1}{2x^2} = \frac{1}{2}x^{-2}$ it is not a polynomial.

(xii) $\frac{1}{\sqrt{5}}x^{\frac{1}{2}} + 1$

This is an expression of x , but power of x is a fraction. Since it is an expression having fractional power of x , so, it is not a polynomial.

(xiii) $\frac{3}{5}x^2 - \frac{7}{3}x + 9$

It is an expression with non-negative integral power of x , having highest power of x is 2.

$\therefore$ it is a polynomial of degree 2.

(xiv) $x^4 - x^{\frac{3}{2}} + x - 3$

This is an expression of x , but power of x is a fraction. Since it is an expression having fractional power of x . So, it is not a polynomial.

(xv) $2x^3 + 3x^2 + \sqrt{x} - 1$

This is an expression of x , but power of x is a fraction. Since it is an expression having fractional power of x . So, it is not a polynomial.

Solution 2:

- (i) $-7 + x$ it is a linear polynomial (polynomial of degree 1)
(ii) $6y$ it is a linear polynomial (polynomial of degree 1)
(iii) $-z^3$ it is a cubic polynomial (polynomial of degree 3)
(iv) $1 - y - y^3$ it is a cubic polynomial (polynomial of degree 3)
(v) $x - x^3 + x^4$ it is a quartic polynomial (polynomial of degree 4)
(vi) $1 + x + x^2$ it is a quadratic polynomial (polynomial of degree 2)
(vii) $-6x^2$ it is a quadratic polynomial (polynomial of degree 2)
(viii) -13 it is a constant polynomial (polynomial of degree 0)
(ix) $-p$ it is a linear polynomial (polynomial of degree 1)

Solution 3:

- (i) the coefficient of x^3 in $x + 3x^2 - 5x^3 + x^4$ is -5 .
(ii) the coefficient of x in $\sqrt{3} - 2\sqrt{2}x + 6x^2$ is $-2\sqrt{2}$.
(iii) the coefficient of x^2 in $2x - 3 + x^3$ is 0 .
(iv) the coefficient of x in $\frac{3}{8}x^2 - \frac{2}{7}x + \frac{1}{6}$ is $-\frac{2}{7}$.
(v) the constant term in $\frac{\pi}{2}x^2 + 7x - \frac{2}{5}\pi$ is $-\frac{2}{5}\pi$

Solution 4:

(i) $\frac{4x - 5x^2 + 6x^3}{2x} = 2 - \frac{5}{2}x + 3x^2$

Degree of the polynomial is 2.

(ii) $y^2(y - y^3) = y^3 - y^5$

Degree of the polynomial is 5.

(iii) $(3x - 2)(2x^3 + 3x^2)$
 $= 6x^4 + 9x^3 - 4x^3 - 6x^2$
 $= 6x^4 + 5x^3 - 6x^2$

Degree of the polynomial is 4.

$$(iv) -\frac{1}{2}x + 3$$

Degree of the polynomial is 1.

$$(v) -8$$

Degree of the polynomial is 0.

$$(vi) x^{-2}(x^4 + x^2)$$

$$= x^2 + x^0$$

$$= x^2 + 1$$

Degree of the polynomial is 2.

Solution 5:

(i) A monomial (polynomial having one term) of degree 5 is x^5

(ii) A binomial (polynomial having two term) of degree 8 is $2x^8 + 2x$

(iii) A trinomial (polynomial having three term) of degree 4 is $2x^4 + 2x + 2$

(iv) A monomial (polynomial having one term) of degree 0 is -2

Solution 6:

$$(i) 5x^3 - 2x^2 + x + 8$$

$$(ii) 2y^3 + 4y^2 - 3y + \frac{2}{3}$$

$$(iii) -x^5 + 6x^3 - 3x^2 + 2x$$

$$(iv) t^4 - 3t^3 - t^2 + t + 2$$

EXERCISE 2B

Solution 1:

$$p(x) = 5 - 4x + 2x^2$$

$$\begin{aligned} \text{(i) } p(0) &= (5 - 4 \times 0 + 2 \times 0^2) \\ &= 5 - 0 + 0 = 5 \end{aligned}$$

$$\begin{aligned} \text{(ii) } p(3) &= (5 - 4 \times 3 + 2 \times 3^2) \\ &= 5 - 12 + 18 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{(iii) } p(-2) &= [5 - 4 \times (-2) + 2 \times (-2)^2] \\ &= (5 + 8 + 8) \\ &= 21 \end{aligned}$$

Solution 2: Given, $p(y) = 4 + 3y - y^2 + 5y^3$

(i) putting $y = 0$, we have

$$\begin{aligned} p(0) &= (4 + 3 \times 0 - 0^2 + 5 \times 0^3) \\ &= (4 + 0 - 0 + 0) \\ &= 4 \end{aligned}$$

(ii) putting $y = 2$, we have

$$\begin{aligned} p(2) &= (4 + 3 \times 2 - 2^2 + 5 \times 2^3) \\ &= 4 + 6 - 4 + 40 \\ &= 46 \end{aligned}$$

(iii) putting $y = (-1)$, we have

$$\begin{aligned} p(-1) &= [4 + 3 \times (-1) - (-1)^2 + 5 \times (-1)^3] \\ &= 4 - 3 - 1 - 5 \\ &= -5 \end{aligned}$$

Solution 3: Given, $f(t) = 4t^2 - 3t + 6$

(i) putting $t = 0$, we have

$$f(0) = (4 \times 0^2 - 3 \times 0 + 6)$$

$$\Rightarrow f(0) = (0 - 0 + 6)$$

$$\Rightarrow f(0) = 6$$

(ii) putting $t = 4$, we have

$$f(4) = (4 \times 4^2 - 3 \times 4 + 6)$$

$$\Rightarrow f(4) = (64 - 12 + 6)$$

$$\Rightarrow f(4) = 58$$

(iii) putting $t = -5$, we have

$$f(-5) = [4 \times (-5)^2 - 3 \times (-5) + 6]$$

$$\Rightarrow f(-5) = (100 + 15 + 6)$$

$$\Rightarrow f(-5) = 121$$

Solution 4:

$$p(x) = x^3 - 3x^2 + 2x$$

$$p(0) = 0^3 - 3 \times 0^2 + 2 \times 0 = 0$$

$$p(1) = 1^3 - 3 \times 1^2 + 2 \times 1 = 1 - 3 + 2 = 0$$

$$p(2) = 2^3 - 3 \times 2^2 + 2 \times 2 = 8 - 3 \times 4 + 4 = 8 - 12 + 4 = 0$$

$\therefore x = 0, 1, 2$ are zeros of the polynomial $p(x)$

Solution 5: Given, $p(x) = x^3 + x^2 - 9x - 9$

(i) Putting $x = 0$

$$p(0) = 0^3 + 0^2 - 9 \times 0 - 9 = -9 \neq 0$$

(ii) Putting $x = 3$

$$p(3) = 3^3 + 3^2 - 9 \times 3 - 9 = 27 + 9 - 27 - 9 = 0$$

(iii) Putting $x = -3$

$$p(-3) = (-3)^3 + (-3)^2 - 9 \times (-3) - 9 = -27 + 9 + 27 - 9 = 0$$

(iv) Putting $x = -1$

$$p(-1) = (-1)^3 + (-1)^2 - 9 \times (-1) - 9 = -1 + 1 + 9 - 9 = 0$$

$\therefore x = 3, -3, -1$ are zeros of $p(x)$ but, $x = 0$ is not zeros of $p(x)$

Solution 6:

(i) $p(x) = x - 4$

For $x = 4$, $p(4) = 4 - 4 = 0$

Hence, 4 is zero of $p(x)$

(ii) $q(x) = x + 3$

For $x = -3$, $q(-3) = -3 + 3 = 0$

Hence, 3 is the zero of $q(x)$

(iii) $f(x) = 2 - 5x$

For $x = \frac{2}{5}$,

$$f\left(\frac{2}{5}\right) = 2 - 5 \times \left(\frac{2}{5}\right) = 2 - 2 = 0$$

Hence, $\frac{2}{5}$ is zero of $f(x)$

(iv) $g(y) = 2y + 1$

For $x = \left(-\frac{1}{2}\right)$,

$$g\left(-\frac{1}{2}\right) = 2 \times \left(-\frac{1}{2}\right) + 1 = -1 + 1$$

Hence, $-\frac{1}{2}$ is the zero of $g(x)$

Solution 7:

(i) $p(x) = x^2 - 3x + 2.$

$$p(1) = 1^2 - 3 \times 1 + 2 = 1 - 3 + 2 = 0$$

$$p(2) = 2^2 - 3 \times 2 + 2 = 4 - 6 + 2 = 0$$

Since $p(1) = 0$ and $p(2) = 0$, hence, 1 and 2 are the zeroes of the given polynomial.

(ii) $q(x) = x^2 + x - 6.$

$$q(2) = 2^2 + 2 - 6 = 4 + 2 - 6 = 0$$

$$q(-3) = (-3)^2 + (-3) - 6 = 9 - 3 - 6 = 0$$

Since $q(2) = 0$ and $q(-3) = 0$, hence, 2 and -3 are the zeroes of the polynomial $q(x)$.

(iii) $r(x) = x^2 - 3x.$

$$r(0) = 0^2 - 3 \times 0 = 0$$

$$r(3) = 3^2 - 3 \times 3 = 9 - 9 = 0$$

Since $r(0) = 0$ and $r(3) = 0$, hence, 0 and 3 are the zeroes of the polynomial $r(x)$.

Solution 8:

(i) $p(x) = x - 5$

Let, $p(x) = 0$

$$\Rightarrow x - 5 = 0$$

$$\Rightarrow x = 5$$

Hence, 5 is the zero of $p(x)$

(ii) $q(x) = x + 4$

Let, $q(x) = 0$

$$\Rightarrow x + 4 = 0$$

$$\Rightarrow x = -4$$

Hence, -4 is the zero of $q(x)$

$$(iii) r(x) = 2x + 5$$

$$\text{Let, } r(x) = 0$$

$$\Rightarrow 2x + 5 = 0$$

$$\Rightarrow 2x = -5$$

$$\Rightarrow x = \frac{-5}{2}$$

Hence, $\frac{-5}{2}$ is zero of the polynomial $r(x)$

$$(iv) f(x) = 3x + 1$$

$$\text{Let, } f(x) = 0$$

$$\Rightarrow 3x + 1 = 0$$

$$\Rightarrow 3x = -1$$

$$\Rightarrow x = -\frac{1}{3}$$

Hence, $x = -\frac{1}{3}$ is zero of $f(x)$

$$(v) g(x) = 5 - 4x$$

$$\text{Let, } g(x) = 0$$

$$\Rightarrow 5 - 4x = 0$$

$$\Rightarrow 4x = 5$$

$$\Rightarrow x = \frac{5}{4}$$

Hence, $x = \frac{5}{4}$ is zero of $g(x)$

$$(vi) h(x) = 6x - 1$$

$$\text{Let, } h(x) = 0$$

$$\Rightarrow 6x - 1 = 0$$

$$\Rightarrow 6x = 1$$

$$\Rightarrow x = \frac{1}{6}$$

Hence, $x = \frac{1}{6}$ is zero of $h(x)$

(vii) $p(x) = ax + b, a \neq 0$

Let, $p(x) = 0$

$$\Rightarrow ax + b = 0$$

$$\Rightarrow ax = -b$$

$$\Rightarrow x = -\frac{b}{a}$$

Hence, $x = -\frac{b}{a}$ is zero of $p(x)$

(viii) $q(x) = 4x$

Let, $q(x) = 0$

$$\Rightarrow 4x = 0,$$

$$\Rightarrow x = 0$$

Hence, $x = 0$ is zero of $q(x)$

(ix) $p(x) = ax, a \neq 0$

Let, $p(x) = 0$

$$\Rightarrow ax = 0,$$

$$\Rightarrow x = 0$$

Hence, $x = 0$ is zero of $p(x)$

Solution 9:

Since, 2 and 0 are the zeros of the polynomial $f(x) = 2x^3 - 5x^2 + ax + b$

$$\therefore f(2) = f(0) = 0$$

Now,

$$f(2) = 0$$

$$\Rightarrow 2 \times 2^3 - 5 \times 2^2 + 2a + b = 0$$

$$\Rightarrow 16 - 20 + 2a + b = 0$$

$$\Rightarrow -4 + 2a + b = 0$$

$$\Rightarrow 2a + b = 4 \quad \dots (i)$$

$$f(0) = 2 \times 0^3 - 5 \times 0^2 + 2 \times 0 + b = 0$$

$$\Rightarrow b = 0 \quad \dots (ii)$$

Putting the value of b in equation (i), we have

$$2a + 0 = 4$$

$$\Rightarrow 2a = 4$$

$$\Rightarrow a = 2$$

$\therefore$ Values of a and b are 2 and 0 respectively

EXERCISE 2C

Solution 1:

$$x^4 + 1 = x^4 + 0.x^3 + 0.x^2 + 0.x + 1$$

$$\begin{array}{r}
 x - 1 \overline{) \begin{array}{l} x^3 + x^2 + x + 1 \\ x^4 + 0.x^3 + 0.x^2 + 0.x + 1 \\ \hline x^3 + 0.x^2 \\ \hline x^3 - x^2 \\ \hline x^2 + 0.x \\ \hline x^2 - x \\ \hline x + 1 \\ \hline x - 1 \\ \hline 2 \end{array}} \\
 \end{array}$$

Quotient is $x^3 + x^2 + x + 1$

Remainder is 2

$$\text{Now, } f(x) = x^4 + 1$$

$$f(1) = 1^4 + 1 = 2$$

$\therefore f(1) = 2$, which is remainder.

Solution 2:

$$\begin{array}{r}
 x + 2 \overline{) \begin{array}{l} 2x^3 - 10x^2 + 22x - 45 \\ 2x^4 - 6x^3 + 2x^2 - x + 2 \\ \hline 2x^4 + 4x^3 \\ \hline -10x^3 + 2x^2 \\ \hline -10x^3 - 20x^2 \\ \hline 22x^2 - x \\ \hline 22x^2 + 44x \\ \hline -45x + 2 \\ \hline -45x - 90 \\ \hline 92 \end{array}} \\
 \end{array}$$

$$\text{Quotient} = 2x^3 - 10x^2 + 22x - 45$$

$$\text{Remainder} = 92$$

We, know

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

$$= (x + 2) \times (2x^3 - 10x^2 + 22x - 45) + 92$$

$$= 2x^4 - 10x^3 + 22x^2 - 45x + 4x^3 - 20x^2 + 44x - 90 + 92$$

$$= 2x^4 - 6x^3 + 2x^2 - x + 2$$

$$= \text{Dividend (Verified)}$$

Solution 3: Given, $p(x) = x^3 - 6x^2 + 9x + 3$ and $g(x) = x - 1$

By remainder theorem, when $p(x)$ is divided by $(x - 1)$, then the remainder = $p(1)$.

$$\text{Now, } p(1) = 1^3 - 6 \times 1^2 + 9 \times 1 + 3$$

$$= 1 - 6 + 9 + 3$$

$$= 7$$

$$\therefore \text{Remainder} = 7$$

Solution 4:

$$\text{Given, } p(x) = 2x^3 - 7x^2 + 9x - 13 \text{ and } g(x) = x - 3$$

By remainder theorem, when $p(x)$ is divided by $g(x) = (x - 3)$, then the remainder = $p(3)$.

$$p(3) = 2 \times 3^3 - 7 \times 3^2 + 9 \times 3 - 13$$

$$= 54 - 63 + 27 - 13$$

$$= 5$$

$$\therefore \text{Remainder} = 5$$

Solution 5:

$$\text{Given, } p(x) = 3x^4 - 6x^2 - 8x - 2 \text{ and } g(x) = x - 2$$

By remainder theorem, when $p(x)$ is divided by $g(x) = (x - 2)$, then the remainder = $p(2)$.

$$p(2) = 3 \times 2^4 - 6 \times 2^2 - 8 \times 2 - 2$$

$$= 6$$

$$\therefore \text{Remainder} = 6$$

Solution 6:

$$p(x) = 2x^3 - 9x^2 + x + 15,$$

$$g(x) = 2x - 3 = 2\left(x - \frac{3}{2}\right)$$

By remainder theorem, when $p(x)$ is divided by $g(x) = 2\left(x - \frac{3}{2}\right)$,

$$\text{Then the remainder} = p\left(\frac{3}{2}\right).$$

$$p\left(\frac{3}{2}\right) = 2 \times \left(\frac{3}{2}\right)^3 - 9 \times \left(\frac{3}{2}\right)^2 + \frac{3}{2} + 15$$

$$= \frac{27}{4} - \frac{81}{4} + \frac{3}{2} + 15$$

$$= \frac{27 - 81 + 6 + 60}{4}$$

$$= \frac{12}{4}$$

$$= 3$$

$$\therefore \text{Remainder} = 3$$

Solution 7:

$$\text{Given, } p(x) = x^3 - 2x^2 - 8x - 1, \text{ and } g(x) = x + 1$$

By remainder theorem, when $p(x)$ is divided by $g(x) = x + 1$,

$$\text{Then the remainder} = p(-1).$$

$$\therefore p(-1) = (-1)^3 - 2 \times (-1)^2 - 8 \times (-1) - 1$$

$$= -1 - 2 + 8 - 1$$

$$= 4$$

$$\text{Remainder} = 4$$

Solution 8:

$$\text{Given, } p(x) = 2x^3 + x^2 - 15x - 12, \text{ and } g(x) = x + 2$$

By remainder theorem, when $p(x)$ is divided by $g(x) = x + 2$,

Then the remainder = $p(-2)$.

$$\begin{aligned}\therefore p(-2) &= 2 \times (-2)^3 + (-2)^2 - 15 \times (-2) - 12 \\ &= -16 + 4 + 30 - 12 \\ &= 6\end{aligned}$$

Remainder = 6

Solution 9:

$$p(x) = 6x^3 + 13x^2 + 3,$$

$$g(x) = 3x + 2$$

$$= 3 \left(x + \frac{2}{3} \right)$$

$$= 3 \left(x - \left(-\frac{2}{3} \right) \right)$$

By remainder theorem, when $p(x)$ is divided by $g(x) = 3x + 2$,

Then the remainder = $p\left(-\frac{2}{3}\right)$.

$$\text{Putting } x = -\frac{2}{3}$$

$$p\left(-\frac{2}{3}\right) = 6 \times \left(-\frac{2}{3}\right)^3 + 13 \times \left(-\frac{2}{3}\right)^2 + 3$$

$$= 6 \times \left(-\frac{8}{27}\right) + 13 \times \frac{4}{9} + 3$$

$$= -\frac{16}{9} + \frac{52}{9} + 3$$

$$= \frac{-16 + 52 + 27}{9}$$

$$= \frac{63}{9}$$

$$= 7$$

Remainder = 7

Solution 10:

$$g(x) = 1 - \frac{3}{2}x = 2 - 3x = -\frac{3}{2}\left(x - \frac{2}{3}\right)$$

By remainder theorem, when $p(x)$ is divided by $g(x) = 1 - \frac{3}{2}x$,

Then the remainder = $p\left(\frac{2}{3}\right)$.

Putting $x = \frac{2}{3}$ in $p(x)$, we get

$$\begin{aligned} p\left(\frac{2}{3}\right) &= \left(\frac{2}{3}\right)^3 - 6 \times \left(\frac{2}{3}\right)^2 + 2 \times \left(\frac{2}{3}\right) - 4 \\ &= \frac{8}{27} - 6 \times \frac{4}{9} + \frac{4}{9} \\ &= \frac{8 - 72 + 36 - 108}{27} \\ &= -\frac{136}{27} \end{aligned}$$

$$\therefore \text{Remainder} = -\frac{136}{27}$$

Solution 11:

$$p(x) = 2x^3 + 3x^2 - 11x - 3,$$

$$g(x) = \left(x + \frac{1}{2}\right)$$

By remainder theorem, when $p(x)$ is divided by $g(x) = x + \frac{1}{2}$,

Then the remainder = $p\left(-\frac{1}{2}\right)$.

Putting, $x = -\frac{1}{2}$

$$\begin{aligned} p\left(-\frac{1}{2}\right) &= 2 \times \left(-\frac{1}{2}\right)^3 + 3 \times \left(-\frac{1}{2}\right)^2 - 11 \times \left(-\frac{1}{2}\right) - 3 \\ &= 2 \times \left(-\frac{1}{8}\right) + \frac{3}{4} + \frac{11}{2} - 3 \\ &= -\frac{1}{4} + \frac{3}{4} + \frac{11}{2} - 3 \end{aligned}$$

$$= \frac{12}{4}$$
$$= 3$$

Solution 12:

$$p(x) = x^3 - ax^2 + 6x - a,$$

$$g(x) = x - a$$

By remainder theorem, when $p(x)$ is divided by $g(x) = x - a$,

Then the remainder = $p(a)$

Putting $x = a$

$$p(a) = a^3 - a \times a^2 + 6 \times a - a$$

$$= a^3 - a^3 + 6a - a$$

$$= 5a$$

$$\therefore \text{Remainder} = 5a$$

Solution 13:

$$\text{Let, } p(x) = 2x^3 + x^2 - ax + 2 \text{ and } q(x) = 2x^3 - 3x^2 - 3x + a$$

When $p(x)$ is divided by $(x - 2)$, then the remainder

$$p(2) = 2 \times 2^3 + 2^2 - 2a + 2$$

$$= 16 + 4 - 2a + 2$$

$$= 22 - 2a$$

When $q(x)$ is divided by $(x - 2)$, then the remainder

$$q(2) = 2 \times 2^3 - 3 \times 2 + a$$

$$= 16 - 12 - 6 + a$$

$$= -2 + a$$

$$\text{Given, } p(x) = q(x)$$

$$\Rightarrow 22 - 2a = -2 + a$$

$$\Rightarrow 3a = 24$$

$$\Rightarrow a = \frac{24}{3} = 8$$

Solution 14:

$$\text{Let } p(x) = x^4 - 2x^3 + 3x^2 - ax + b$$

When $p(x)$ is divided by $(x - 1)$, then the remainder

$$\begin{aligned} p(1) &= 1^4 - 2 \times 1^3 + 3 \times 1^2 - a \times 1 + b \\ &= 2 - a + b \end{aligned}$$

When $p(x)$ is divided by $(x + 1)$, then the remainder

$$\begin{aligned} p(-1) &= (-1)^4 - 2 \times (-1)^3 + 3 \times (-1)^2 - a \times (-1) + b \\ &= 1 + 2 + 3 + a + b \\ &= 6 + a + b \end{aligned}$$

$$\text{Since, } p(1) = 5$$

$$\Rightarrow 2 - a + b = 5 \quad \dots (1)$$

$$p(-1) = 19$$

$$\Rightarrow 6 + a + b = 19 \quad \dots (2)$$

Adding (1) and (2), we get

$$2 + 6 - a + b + a + b = 5 + 19$$

$$\Rightarrow 8 + 2b = 24$$

$$\Rightarrow 2b = 16$$

$$\Rightarrow b = 8$$

Now, from (2)

$$6 + a + 8 = 19$$

$$\Rightarrow a = 19 - 6 - 8 = 5$$

$$\therefore a = 5, b = 8$$

When $p(x)$ is divided by $(x - 2)$, then the remainder is

$$\begin{aligned} p(2) &= 2^4 - 2 \times 2^3 + 3 \times 2^2 - 2a + b \\ &= 16 - 16 + 12 - 2 \times 5 + 8 \quad [\because a = 5, b = 8] \\ &= 10 \end{aligned}$$

Solution 15:

$$p(x) = x^3 - 5x^2 + 4x - 3$$

$$g(x) = x - 2$$

Putting $x = 2$

$$\begin{aligned} p(2) &= 2^3 - 5 \times 2^2 + 4 \times 2 - 3 \\ &= 8 - 20 + 8 - 3 \\ &= -7 \end{aligned}$$

$$\therefore p(2) \neq 0$$

By factor theorem, $(x - 2)$ is not a factor of $p(x)$.

Hence, $p(x)$ is not a multiple of $g(x)$.

Solution 16:

$$p(x) = 2x^3 - 11x^2 - 4x + 5$$

$$g(x) = 2x + 1 = 2\left(x + \frac{1}{2}\right) = 2\left[x - \left(-\frac{1}{2}\right)\right]$$

$$\text{Putting } x = -\frac{1}{2}$$

$$\begin{aligned} p\left(-\frac{1}{2}\right) &= 2 \times \left(-\frac{1}{2}\right)^3 - 11 \times \left(-\frac{1}{2}\right)^2 - 4 \times \left(-\frac{1}{2}\right) + 5 \\ &= -\frac{1}{4} - \frac{11}{4} + 2 + 5 \\ &= -\frac{12}{4} + 7 = -3 + 7 = 4 \end{aligned}$$

$$\therefore p\left(-\frac{1}{2}\right) \neq 0$$

By factor theorem, $g(x) = 2x + 1$, is not a factor of $p(x)$.

Hence, $p(x)$ is not a multiple of $g(x)$.

EXERCISE 2D

Solution 1:

Let,

$$p(x) = x^3 - 8$$

By the factor theorem, $(x - 2)$ is a factor of $(x^3 - 8)$ if $p(2) = 0$

$$p(2) = 2^3 - 8$$

$$= 8 - 8$$

$$= 0$$

$\therefore (x - 2)$ is a factor of $(x^3 - 8)$

Solution 2:

Let, $p(x) = 2x^3 + 7x^2 - 24x - 45$

By the factor theorem, $(x - 3)$ is a factor of $p(x)$ if $p(3) = 0$.

$$p(3) = 2 \times 3^3 + 7 \times 3^2 - 24 \times 3 - 45$$

$$= 54 + 63 - 72 - 45$$

$$= 0$$

$\therefore (x - 3)$ is a factor of $2x^3 + 7x^2 - 24x - 45$

Solution 3:

Let, $p(x) = (2x^4 + 9x^3 + 6x^2 - 11x - 6)$

By the factor theorem, $(x - 1)$ is a factor of $p(x)$ if $p(1) = 0$.

Now,

$$p(1) = 2 \times 1^4 + 9 \times 1^3 + 6 \times 1^2 - 11 \times 1 - 6$$

$$= 2 + 9 + 6 - 11 - 6$$

$$= 0$$

$\therefore (x - 1)$ is a factor of $p(x)$

Solution 4:

Let, $p(x) = (x^4 - x^2 - 12)$

$$g(x) = x + 2$$

By the factor theorem, $(x + 2)$ is a factor of $p(x)$ if $p(-2) = 0$.

Now,

$$\begin{aligned} p(-2) &= (-2)^4 - (-2)^2 - 12 \\ &= 16 - 4 - 12 \\ &= 0 \end{aligned}$$

$\therefore (x + 2)$ is a factor of $x^4 - x^2 - 12$

Solution 5:

$$p(x) = 69 + 11x - x^2 + x^3$$

$$g(x) = x + 3$$

By the factor theorem, $(x + 3)$ is a factor of $p(x)$ if $p(-3) = 0$.

Putting $x = -3$ in $p(x)$

$$\begin{aligned} p(-3) &= 69 + 11 \times (-3) - (-3)^2 - (-3)^3 \\ &= 69 - 33 - 9 - 27 \\ &= 0 \end{aligned}$$

$\therefore g(x) = (x + 3)$ is a factor of $p(x) = 69 + 11x - x^2 + x^3$.

Solution 6:

$$\text{Let, } p(x) = 2x^3 + 9x^2 - 11x - 30$$

By the factor theorem, $(x + 5)$ is a factor of $p(x)$ if $p(-5) = 0$.

Now, Putting $x = -5$ in $p(x)$

$$\begin{aligned} p(-5) &= 2 \times (-5)^3 + 9 \times (-5)^2 - 11 \times (-5) - 30 \\ &= -250 + 225 + 55 - 30 \\ &= 0 \end{aligned}$$

$\therefore (x + 5)$ is a factor of $2x^3 + 9x^2 - 11x - 30$

Solution 7:

$$p(x) = 2x^4 + x^3 - 8x^2 - x + 6$$

$$g(x) = 2x - 3 = 2\left(x - \frac{3}{2}\right)$$

By the factor theorem, $(2x - 3)$ is a factor of $p(x)$ if $p\left(\frac{3}{2}\right) = 0$.

Now, Putting $x = \frac{3}{2}$ in $p(x)$

$$\begin{aligned} p\left(\frac{3}{2}\right) &= 2 \times \left(\frac{3}{2}\right)^4 + \left(\frac{3}{2}\right)^3 - 8 \times \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right) + 6 \\ &= \frac{81}{8} + \frac{27}{8} - 18 - \frac{3}{2} + 6 \\ &= \frac{81 + 27 - 144 - 12 + 48}{8} \\ &= \frac{0}{8} \\ &= 0 \end{aligned}$$

$\therefore (2x - 3)$ is a factor of $2x^4 + x^3 - 8x^2 - x + 6$

Solution 8:

$$p(x) = 3x^3 + x^2 - 20x + 12$$

$$g(x) = 3x - 2 = 3\left(x - \frac{2}{3}\right)$$

By the factor theorem, $(3x - 2)$ is a factor of $p(x)$ if $p\left(\frac{2}{3}\right) = 0$.

Now, Putting $x = \frac{2}{3}$ in $p(x)$

$$\begin{aligned} P\left(\frac{2}{3}\right) &= 3 \times \left(\frac{2}{3}\right)^3 + \left(\frac{2}{3}\right)^2 - 20 \times \frac{2}{3} + 12 \\ &= \frac{8}{9} + \frac{4}{9} - \frac{40}{3} + 12 \\ &= \frac{8 + 4 - 120 + 108}{9} \\ &= \frac{120 - 120}{9} \\ &= 0 \end{aligned}$$

$\therefore g(x) = (3x - 2)$ is a factor of $p(x) = 3x^3 + x^2 - 20x + 12$

Solution 9:

$$\text{Let, } p(x) = (7x^2 - 4\sqrt{2}x - 6)$$

$$g(x) = (x - \sqrt{2})$$

By the factor theorem, $(x - \sqrt{2})$ is a factor of $p(x)$ if $p(\sqrt{2}) = 0$.

Now, Putting $x = \sqrt{2}$ in $p(x)$

$$p(\sqrt{2}) = 7 \times (\sqrt{2})^2 - 4 \times \sqrt{2} \times \sqrt{2} - 6$$

$$= 14 - 8 - 6$$

$$= 0$$

$\therefore g(x) = (x - \sqrt{2})$ is a factor of $p(x) = (7x^2 - 4\sqrt{2}x - 6)$

Solution 10:

$$\text{Let, } p(x) = (2\sqrt{2}x^2 + 5x + \sqrt{2})$$

$$g(x) = (x + \sqrt{2})$$

By the factor theorem, $(x + \sqrt{2})$ is a factor of $p(x)$ if $p(-\sqrt{2}) = 0$.

Now, Putting $x = -\sqrt{2}$ in $p(x)$

$$p(-\sqrt{2}) = 2\sqrt{2} \times (-\sqrt{2})^2 + 5 \times (-\sqrt{2}) + \sqrt{2}$$

$$= 4\sqrt{2} - 5\sqrt{2} + \sqrt{2}$$

$$= 0$$

$\therefore (x + \sqrt{2})$ is a factor of $(2\sqrt{2}x^2 + 5x + \sqrt{2})$

Solution 11:

$$\text{Let, } f(p) = p^{10} - 1$$

$$g(p) = p^{11} - 1$$

Putting $p = 1$ in $f(p)$ and $g(p)$

$$f(1) = 1^{10} - 1$$

$$= 1 - 1$$

$$= 0,$$

$$\begin{aligned}g(1) &= 1^{11} - 1 \\&= 1 - 1 \\&= 0\end{aligned}$$

By the factor theorem, $(p - 1)$ is a factor of $(p^{10} - 1)$ and $(p^{11} - 1)$

Solution 12:

$$\text{Let, } p(x) = (2x^3 + 9x^2 + x + k)$$

$$g(x) = (x - 1)$$

Since, $g(x) = (x - 1)$ is a factor of $p(x)$

$$\therefore p(1) = 0$$

$$\Rightarrow 2 \times 1^3 + 9 \times 1^2 + 1 + k = 0$$

$$\Rightarrow 12 + k = 0$$

$$\Rightarrow k = -12$$

$\therefore$ The value of $k = -12$

Solution 13:

$$\text{Let, } p(x) = 2x^3 - 3x^2 - 18x + a$$

Since, $(x - 4)$ is a factor of $p(x) = 2x^3 - 3x^2 - 18x + a$

$$\therefore p(4) = 0$$

$$\Rightarrow 2 \times 4^3 - 3 \times 4^2 - 18 \times 4 + a = 0$$

$$\Rightarrow 128 - 48 - 72 + a = 0$$

$$\Rightarrow 8 + a = 0$$

$$\Rightarrow a = -8$$

$\therefore$ The value of $a = -8$

Solution 14:

$$\text{Let, } p(x) = (ax^3 + x^2 - 2x + 4a - 9)$$

Since, $(x + 1)$ is a factor of $p(x) = (ax^3 + x^2 - 2x + 4a - 9)$

$$\therefore p(-1) = 0$$

$$\Rightarrow a \times (-1)^3 + (-1)^2 - 2 \times (-1) + 4a - 9 = 0$$

$$\Rightarrow -a + 1 + 2 + 4a - 9 = 0$$

$$\Rightarrow 3a - 6 = 0$$

$$\Rightarrow 3a = 6$$

$$\Rightarrow a = \frac{6}{3} = 2$$

$\therefore$ The value of $a = 2$

Solution 15:

$$\text{Let, } p(x) = (x^5 - 4a^2x^3 + 2x + 2a + 3)$$

Since, $(x + 2a)$ is a factor of $(x^5 - 4a^2x^3 + 2x + 2a + 3)$

$$p(-2a) = 0$$

$$\Rightarrow (-2a)^5 - 4a^2 \times (-2a)^3 + 2 \times (-2a) + 2a + 3 = 0$$

$$\Rightarrow -32a^5 + 32a^5 - 4a + 2a + 3 = 0$$

$$\Rightarrow -2a + 3 = 0$$

$$\Rightarrow a = \frac{3}{2}$$

$\therefore$ The value of $a = \frac{3}{2}$

Solution 16:

$$\text{Let, } p(x) = (8x^4 + 4x^3 - 16x^2 + 10x + m)$$

$$g(x) = 2x - 1 = 2\left(x - \frac{1}{2}\right)$$

Since, $(2x - 1)$ is a factor of $(8x^4 + 4x^3 - 16x^2 + 10x + m)$

$$\therefore p\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow 8 \times \left(\frac{1}{2}\right)^4 + 4 \times \left(\frac{1}{2}\right)^3 - 16 \times \left(\frac{1}{2}\right)^2 + 10 \times \frac{1}{2} + m = 0$$

$$\Rightarrow \frac{8}{16} + \frac{4}{8} - \frac{16}{4} + \frac{10}{2} + m = 0$$

$$\Rightarrow 2 + m = 0$$

$$\Rightarrow m = -2$$

$\therefore$ The value of $m = -2$

Solution 17:

$$\text{Let, } p(x) = (x^4 - x^3 - 11x^2 - x + a)$$

Since, $(x + 3)$ is a factor of $(x^4 - x^3 - 11x^2 - x + a)$ then $p(-3) = 0$

$$p(-3) = 0$$

$$\Rightarrow (-3)^4 - (-3)^3 - 11 \times (-3)^2 - (-3) + a = 0$$

$$\Rightarrow 81 + 27 - 11 \times 9 + 3 + a = 0$$

$$\Rightarrow 81 + 27 - 99 + 3 + a = 0$$

$$\Rightarrow 12 + a = 0$$

$$\Rightarrow a = -12$$

$\therefore p(x) = (x^4 - x^3 - 11x^2 - x + a)$ is divisible by $(x + 3)$ when $a = -12$

Solution 18:

$$\text{Let, } p(x) = x^3 - 3x^2 - 13x + 15$$

$$g(x) = x^2 + 2x - 3$$

$$= x^2 + 3x - x - 3$$

$$= x(x + 3) - (x + 3)$$

$$= (x + 3)(x - 1)$$

Now, $p(x)$ will be exactly divisible by $g(x)$ if it is exactly divisible by both $(x - 1)$ and $(x + 3)$.

So, from factor theorem, we have $p(1) = 0$ and $p(-3) = 0$

Now,

$$p(1) = 1^3 - 3 \times 1^2 - 13 \times 1 + 15$$

$$= 1 - 3 - 13 + 15$$

$$= 0$$

And,

$$\begin{aligned}p(-3) &= (-3)^3 - 3 \times (-3)^2 - 13 \times (-3) + 15 \\&= (-27 - 27 + 39 + 15) \\&= 0\end{aligned}$$

Since, $p(1) = 0$ and $p(-3) = 0$

$\therefore p(x) = x^3 - 3x^2 - 13x + 15$ is exactly divisible by $x^2 + 2x - 3$

Solution 19:

$$\text{Let, } p(x) = (x^3 + ax^2 + bx + 6)$$

Since, $(x - 2)$ is a factor of $p(x)$

$$\therefore p(2) = 0$$

$$\Rightarrow 2^3 + a \times 2^2 + b \times 2 + 6 = 0$$

$$\Rightarrow 8 + 4a + 2b + 6 = 0$$

$$\Rightarrow 14 + 4a + 2b = 0$$

$$\Rightarrow 2a + b = -7 \quad \dots (i)$$

Now, by the factor theorem, we can say:

When $p(x)$ will be divided by $(x - 3)$, 3 will be its remainder.

$$\Rightarrow p(3) = 3$$

Now,

Putting $x = 3$

$$p(3) = 3^3 + a \times 3^2 + b \times 3 + 6$$

$$= 27 + 9a + 3b + 6$$

$$= 33 + 9a + 3b$$

$$\text{Since, } p(3) = 3$$

$$\Rightarrow 33 + 9a + 3b = 3$$

$$\Rightarrow 9a + 3b = -30$$

$$\Rightarrow 3a + b = -10 \quad \dots (ii)$$

Subtracting (i) from (ii), we get

$$3a + b - 2a - b = -10 - (-7)$$

$$\Rightarrow a = -3$$

Putting $a = -3$ in equation (ii) we get (ii)

$$3(-3) + b = -10$$

$$\Rightarrow -9 + b = -10$$

$$\Rightarrow b = -1$$

Solution 20:

$$\text{Let, } p(x) = x^3 - 10x^2 + ax + b$$

By factor theorem $p(x) = (x^3 - 10x^2 + ax + b)$ is divisible by $(x - 1)$ if $p(1) = 0$

$$\therefore p(1) = 0$$

$$\Rightarrow 1^3 - 10 \times 1^2 + a \times 1 + b = 0$$

$$\Rightarrow 1 - 10 + a + b = 0$$

$$\Rightarrow -9 + a + b = 0$$

$$\Rightarrow a + b = 9 \quad \dots (1)$$

By factor theorem $p(x) = (x^3 - 10x^2 + ax + b)$ is divisible by $(x - 2)$ if $p(2) = 0$

$$\therefore p(2) = 0$$

$$\Rightarrow 2^3 - 10 \times 2^2 + a \times 2 + b = 0$$

$$\Rightarrow 8 - 40 + 2a + b = 0$$

$$\Rightarrow -32 + 2a + b = 0$$

$$\Rightarrow 2a + b = 32 \quad \dots (2)$$

Subtracting (1) from (2), we get

$$2a + b - (a + b) = 32 - 9$$

$$a = 23$$

From (2), we get

$$2 \times 23 + b = 32$$

$$\Rightarrow 46 + b = 32$$

$$\Rightarrow b = 32 - 46$$

$$\Rightarrow b = -14$$

The value of a and b are 23 and -14

Solution 21:

$$\text{Let, } p(x) = (x^4 + ax^3 - 7x^2 - 8x + b)$$

By factor theorem $p(x) = (x^4 + ax^3 - 7x^2 - 8x + b)$ is divisible by $(x + 2)$

$$\text{if } p(-2) = 0$$

Now, Putting $x = -2$

$$p(-2) = (-2)^4 + a \times (-2)^3 - 7 \times (-2)^2 - 8 \times (-2) + b$$

$$= 16 - 8a - 28 + 16 + b$$

$$= 4 - 8a + b$$

$$\therefore 4 - 8a + b = 0$$

$$\Rightarrow 8a - b = 4 \quad \dots (1)$$

Also,

By factor theorem $p(x) = (x^4 + ax^3 - 7x^2 - 8x + b)$ is divisible by $(x + 3)$

$$\text{if } p(-3) = 0$$

Now, Putting $x = -3$

$$p(-3) = (-3)^4 + a \times (-3)^3 - 7 \times (-3)^2 - 8 \times (-3) + b$$

$$= 81 - 27a - 63 + 24 + b$$

$$= 42 - 27a + b$$

$$\Rightarrow 42 - 27a + b = 0$$

$$\Rightarrow 27a - b = 42 \quad \dots (2)$$

Subtracting (1) from (2), we get

$$\Rightarrow 19a = 38$$

$$\Rightarrow a = 2$$

Putting $a = 2$, in equation (2), we get

$$27 \times 2 - b = 42$$

$$\Rightarrow 54 - b = 42$$

$$\Rightarrow b = 12$$

Solution 22:

$$\text{Let, } p(x) = px^2 + 5x + r$$

Since, $(x - 2)$ and $\left(x - \frac{1}{2}\right)$ are factors of $px^2 + 5x + r$.

$$\therefore p(2) = 0, \text{ and } p\left(\frac{1}{2}\right) = 0$$

Now,

$$p(2) = 0$$

$$\Rightarrow p \times 2^2 + 5 \times 2 + r = 0$$

$$\Rightarrow 4p + r = -10 \quad \dots (1)$$

Also,

$$p\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow p \times \left(\frac{1}{2}\right)^2 + 5 \times \frac{1}{2} + r = 0$$

$$\Rightarrow p \times \frac{1}{4} + \frac{5}{2} + r = 0$$

$$\Rightarrow \frac{p}{4} + r = -\frac{5}{2}$$

$$\Rightarrow p + 4r = -10 \quad \dots (2)$$

From (1) and (2), we get

$$4p + r = p + 4r$$

$$\Rightarrow 4p - p = 4r - r$$

$$\Rightarrow 3p = 3r$$

$$\therefore p = r$$

Solution 23:

$$\text{Let, } p(x) = 2x^4 - 5x^3 + 2x^2 - x + 2$$

$$g(x) = x^2 - 3x + 2$$

$$= x^2 - 2x - x + 2$$

$$= x(x - 2) - (x - 2)$$

$$= (x - 2)(x - 1)$$

Now,

$p(x)$ will be divisible by $g(x)$ if $p(x)$ is exactly divisible by both $(x - 1)$ and $(x - 2)$.

By factor theorem $p(x) = 2x^4 - 5x^3 + 2x^2 - x + 2$ is divisible by $(x - 2)$ and $(x - 1)$

if $p(2) = 0$ and $p(1) = 0$

Putting $x = 2$

$$p(2) = 2 \times 2^4 - 5 \times 2^3 + 2 \times 2^2 - 2 + 2$$

$$= 32 - 40 + 8 - 2 + 2$$

$$= 0$$

Also, putting $x = 1$

$$p(1) = 2 \times 1^4 - 5 \times 1^3 + 2 \times 1^2 - 1 + 2$$

$$= 2 - 5 + 2 - 1 + 2$$

$$= 0$$

$$\therefore p(1) = p(2) = 0$$

$\therefore p(x) = 2x^4 - 5x^3 + 2x^2 - x + 2$ is divisible by both $(x - 1)$ and $(x - 2)$.

Hence, $2x^4 - 5x^3 + 2x^2 - x + 2$ is exactly divisible by $x^2 - 3x + 2$.

Solution 24:

Let $p(x) = 2x^4 - 5x^3 + 2x^2 - x - 3$ and m be added to $p(x)$ so that the results is exactly divisible by $(x - 2)$,

$\therefore p(x) = 2x^4 - 5x^3 + 2x^2 - x - 3 + m$ is exactly divisible by $(x - 2)$.

By factor theorem $p(x) = 2x^4 - 5x^3 + 2x^2 - x - 3 + m$ is divisible by $(x - 2)$ if $p(2) = 0$

Now, Putting $x = 2$

$$p(2) = 0$$

$$\Rightarrow 2 \times 2^4 - 5 \times 2^3 + 2 \times 2^2 - 2 - 3 + m = 0$$

$$\Rightarrow 32 - 40 + 8 - 5 + m = 0$$

$$\Rightarrow -8 + 8 - 5 + m = 0$$

$$\Rightarrow m = 5$$

$\therefore 5$ must be added to $2x^4 - 5x^3 + 2x^2 - x - 3$ so that the result is exactly divisible by $(x - 2)$.

Solution 25:

$$\text{Let } p(x) = x^4 + 2x^3 - 2x^2 + 4x + 6$$

$$q(x) = x^2 + 2x - 3$$

$$r(x) = ax + b$$

$r(x)$ be Subtracted from $p(x)$ so that the results exactly divisible by $q(x)$

$\therefore p(x) = x^4 + 2x^3 - 2x^2 + 4x + 6 - (ax + b)$ is exactly divisible by $q(x)$

By factor theorem $p(x) = x^4 + 2x^3 - 2x^2 + 4x + 6 - (ax + b)$ is divisible by $q(x)$

Now,

$$p(x) = x^4 + 2x^3 - 2x^2 + 4x + 6 - (ax + b)$$

$$\Rightarrow p(x) = x^4 + 2x^3 - 2x^2 + (4 - a)x + 6 - b \quad \dots(i)$$

$$q(x) = x^2 + 2x - 3$$

$$\Rightarrow q(x) = x^2 + 3x - x - 3$$

$$\Rightarrow q(x) = x(x + 3) - (x + 3)$$

$$\Rightarrow q(x) = (x + 3)(x - 1)$$

$\therefore (x + 3)$ and $(x - 1)$ are the factors of $p(x)$ if $p(1) = p(-3) = 0$

Putting $x = 1$ in (i), we get

$$p(1) = 1^4 + 2 \times 1^3 - 2 \times 1^2 + (4 - a) + 6 - b = 0$$

$$\Rightarrow 1 + 2 - 2 + 4 - a + 6 - b = 0$$

$$\Rightarrow 11 - a - b = 0$$

$$\Rightarrow a + b = 11 \quad \dots(1)$$

Also, putting $x = -3$ in (i), we get

$$p(-3) = 0$$

$$\Rightarrow (-3)^4 + 2 \times (-3)^3 - 2 \times (-3)^2 + (4 - a)(-3) + 6 - b = 0$$

$$\Rightarrow 81 - 54 - 18 - 12 + 3a + 6 - b = 0$$

$$\Rightarrow 3 + 3a - b = 0$$

$$\Rightarrow 3a - b = -3 \quad \dots (2)$$

By subtracting both the equations (1) and (2)

$$-a - 3a - b + b = -11 + 3$$

$$\Rightarrow -4a = -8$$

$$\Rightarrow a = 2$$

From equation (2)

$$3 \times 2 - b = -3$$

$$\Rightarrow 6 - b = -3$$

$$\Rightarrow b = 9$$

$$\therefore r(x) = ax + b = 2 \times x + 9 = 2x + 9$$

Thus, the remainder $(2x + 9)$ must be subtracted from $(x^4 + 2x^3 - 2x^2 + 4x + 6)$ so that the result is exactly divisible by $(x^2 + 2x - 3)$.

Solution 26:

$$\text{Let, } p(x) = x^n + a^n$$

$$\text{Putting } x = -a$$

$$p(-a) = (-a)^n + a^n$$

If n is any odd positive integer, then

$$p(-a) = (-a)^n + a^n = -a^n + a^n = 0$$

Therefore, by factor theorem, $(x + a)$ is a factor of $(x^n + a^n)$ for any odd positive integer.

MCQs

Answer 1: (c) $\sqrt{2}x^2 - \sqrt{3}x + 6$

Answer 2: (d) $x^2 + \frac{2x^{\frac{3}{2}}}{\sqrt{x}} + 6$

$$x^2 + \frac{2x^{\frac{3}{2}}}{\sqrt{x}} + 6$$

$$= x^2 + 2x^{\frac{3}{2}} \cdot x^{-\frac{1}{2}} + 6$$

$$= x^2 + 2x + 6$$

$\therefore$ It is a polynomial.

Answer 3: (c) y

y is a polynomial because it has a non-negative integral power 1.

Answer 4: (d) -4

-4 is a constant polynomial of degree zero.

Answer 5: (d) 0

0 is a polynomial whose degree is not defined.

Answer 6: (d) $x^2 + 5x + 4$

$x^2 + 5x + 4$ is a polynomial of degree 2. So, it is a quadratic polynomial.

Answer 7: (b) $x + 1$

Since, $x + 1$ is a polynomial of degree 1.

Answer 8: (b) $x^2 + 4$

This is an expression having two non-zero terms. So, it is a binomial.

Answer 9: (d) 0 $\sqrt{3}$ is a constant polynomial of degree 0.**Answer 10:** (c) not defined**Answer 11:** (d) not defined**Answer 12:** (d) 8

$$p(x) = x + 4$$

$$p(-x) = (-x) + 4$$

$$\therefore p(x) + p(-x) = x + 4 - x + 4 = 8$$

Answer 13: (b) 1

$$p(x) = x^2 - 2\sqrt{2}x + 1$$

$$\begin{aligned}\therefore p(2\sqrt{2}) &= (2\sqrt{2})^2 - 2\sqrt{2} \times (2\sqrt{2}) + 1 \\ &= (2^2 \times \sqrt{2}^2) - 8 + 1 \\ &= 8 - 8 + 1 \\ &= 1\end{aligned}$$

Answer 14: (d) -6

$$p(x) = 5x - 4x^2 + 3$$

Putting $x = -1$

$$p(-1) = 5 \times (-1) - 4 \times (-1)^2 + 3 = -5 - 4 + 3 = -6$$

 $\therefore$ (d) is the correct answer.**Answer 15:** (d) 50

$$\text{Let } f(x) = x^{51} + 51$$

By remainder theorem, when $f(x)$ is divided by $(x + 1)$, then the remainder $= f(-1)$.

Putting $x = -1$

$$\begin{aligned}f(-1) &= (-1)^{51} + 51 \\&= -1 + 51 \\&= 50\end{aligned}$$

$\therefore$ Remainder = 50

Hence, (d) is the correct answer.

Answer 16: (c) 2

$\therefore (x + 1)$ is a factor of $2x^2 + kx$, $p(-1) = 0$

Putting $x = -1$

$$p(-1) = 2 \times (-1)^2 + k \times (-1) = 0$$

$$\Rightarrow 2 - k = 0$$

$$\Rightarrow k = 2$$

(c) is correct answer.

Answer 17:

$$p(x) = x^4 + 2x^3 - 3x^2 + x - 1$$

Since, $p(x)$ is divided by $(x - 2)$, so the remainder is $p(2)$

$$\text{Now, } p(2) = 2^4 + 2 \times 2^3 - 3 \times 2^2 + 2 - 1$$

$$= 16 + 16 - 12 + 1$$

$$= 21$$

Answer 18: (d) 4

Since, $p(x) = x^3 - 3x^2 + 4x + 32$ is divisible by $(x + 2)$. By remainder theorem $p(-2)$ is remainder.

$$\text{Now, } p(-2) = (-2)^3 - 3 \times (-2)^2 + 4 \times (-2) + 32$$

$$= -8 - 12 - 8 + 32$$

$$= 4$$

(d) is correct answer.

Answer 19: (c) -2

$$2x - 1 = 0$$

$$\Rightarrow x = \frac{1}{2}$$

By the remainder theorem, when $p(x)$ is divided by $(2x - 1)$, the remainder is $p\left(\frac{1}{2}\right)$.

Now,

$$\begin{aligned} p\left(\frac{1}{2}\right) &= 4 \times \left(\frac{1}{2}\right)^3 - 12 \times \left(\frac{1}{2}\right)^2 + 11 \times \frac{1}{2} - 5 \\ &= \frac{1}{2} - 3 + \frac{11}{2} - 5 \\ &= -2 \end{aligned}$$

Answer 20: (b) a

By remainder theorem, when $p(x) = x^3 - ax^2 + x$ is divided by $(x - a)$, then the remainder is $p(a)$.

Putting $x = a$

$$p(a) = a^3 - a \times a^2 + a = a$$

$\therefore$ Remainder is a

(b) is correct answer.

Answer 21: (c) $-a$

By remainder theorem, when $p(x) = x^3 + ax^2 + 2x + a$ is divided by $(x + a)$, then the remainder is $p(-a)$.

$$\begin{aligned} \therefore p(-a) &= (-a)^3 + a \times (-a)^2 + 2 \times (-a) + a \\ &= -a^3 + a^3 - 2a + a \\ &= -a \end{aligned}$$

$\therefore$ (c) is correct.

Answer 22: (d) $x^3 - 2x^2 - x - 2$

$$x + 1 = 0$$

$$\Rightarrow x = -1$$

Given,

$$p(x) = x^3 - 2x^2 - x - 2$$

$$\begin{aligned} p(-1) &= (-1)^3 - 2 \times (-1)^2 - (-1) + 2 \\ &= 0 \end{aligned}$$

$\therefore (x + 1)$ is a factor of $x^3 - 2x^2 - x - 2$

Answer 23: (b) $-\frac{5}{2}$

$$p(x) = 2x + 5$$

Now, zeros of the polynomial is obtained by

$$p(x) = 0$$

$$\Rightarrow 2x + 5 = 0$$

$$\Rightarrow x = -\frac{5}{2}$$

(b) is correct answer.

Answer 24:

$$p(x) = x^2 + x - 6$$

Now, zeros of the polynomial is obtained by

$$p(x) = 0$$

$$\Rightarrow x^2 + x - 6 = 0$$

$$\Rightarrow x^2 + 3x - 2x - 6 = 0$$

$$\Rightarrow x(x + 3) - 2(x + 3) = 0$$

$$\Rightarrow (x + 3)(x - 2) = 0$$

$$\therefore x = 2, -3$$

(c) is correct answer.

Answer 25: (b) $\frac{1}{2}, -3$

$$p(x) = 2x^2 + 5x - 3$$

Now, zeros of the polynomial is obtained by

$$p(x) = 0$$

$$\Rightarrow 2x^2 + 5x - 3 = 0$$

$$\Rightarrow 2x^2 + 6x - x - 3 = 0$$

$$\Rightarrow 2x(x + 3) - (x + 3) = 0$$

$$\Rightarrow (x + 3)(2x - 1) = 0$$

$$\therefore x = \frac{1}{2}, -3$$

(b) is correct answer.

Answer 26: (c) $-4, \frac{1}{2}$

$$p(x) = 2x^2 + 7x - 4$$

Now, zeros of the polynomial is obtained by

$$p(x) = 0$$

$$\Rightarrow 2x^2 + 7x - 4 = 0$$

$$\Rightarrow 2x^2 + 8x - x - 4 = 0$$

$$\Rightarrow 2x(x + 4) - (x + 4) = 0$$

$$\Rightarrow (x + 4)(2x - 1) = 0$$

$$\therefore x = -4, \frac{1}{2}$$

(c) is correct answer.

Answer 27: (b) 5

Since, $(x + 5)$ is a factor of $p(x) = x^3 - 20x + 5k$,

$$\therefore p(-5) = 0$$

$$\Rightarrow (-5)^3 - 20 \times (-5) + 5k = 0$$

$$\Rightarrow -125 + 100 + 5k = 0$$

$$5k = 25$$

$$\Rightarrow k = 5$$

(b) is correct answer.

Answer 28: (b) $m = 7, n = -18$

$$\text{Let, } p(x) = (x^3 + 10x^2 + mx + n)$$

Since, $(x + 2)$ and $(x - 1)$ are factors of $(x^3 + 10x^2 + mx + n)$,

$$\therefore p(-2) = 0 \text{ and } p(1) = 0$$

Now, Putting $x = -2$

$$p(-2) = (-2)^3 + 10 \times (-2)^2 - 2m + n$$

$$= -8 + 40 - 2m + n$$

$$= 32 - 2m + n$$

$$\therefore 32 - 2m + n = 0$$

$$\Rightarrow 2m - n = 32 \quad \dots (1)$$

Also, $p(1) = 0$

$$\Rightarrow 1^3 + 10 \times 1^2 + m + n = 0$$

$$\Rightarrow 11 + m + n = 0$$

$$\Rightarrow m + n = -11 \quad \dots (2)$$

Adding (1) and (2), we get

$$2m - n + m + n = 32 - 11$$

$$\Rightarrow 3m = 21$$

$$m = 7$$

From (1), we get

$$2 \times 7 - n = 32$$

$$\Rightarrow 14 - n = 32$$

$$\Rightarrow n = -18$$

Answer 29: (a) 1

$$\text{Let, } p(x) = (x^{100} + 2x^{99} + k)$$

Since, $p(x)$ is divisible by $(x + 1)$, $p(-1) = 0$

Now,

$$p(-1) = 0$$

$$\Rightarrow (-1)^{100} + 2 \times (-1)^{99} + k = 0$$

$$\Rightarrow 1 - 2 + k = 0$$

$$\Rightarrow -1 + k = 0$$

$$\Rightarrow k = 1$$

Answer 30: (d) -3

$$\text{Let, } p(x) = 2x^3 - kx^2 + 3x + 10$$

Since, $p(x) = 2x^3 - kx^2 + 3x + 10$ exactly divisible by $(x + 2)$

$$\therefore p(-2) = 0$$

$$\Rightarrow 2 \times (-2)^3 - k \times (-2)^2 + 3 \times (-2) + 10 = 0$$

$$\Rightarrow -16 - 4k - 6 + 10 = 0$$

$$\Rightarrow -12 - 4k = 0$$

$$\Rightarrow 4k = -12$$

$$\Rightarrow k = -3$$

Answer 31: (b) 0, 3

$$\text{Let, } p(x) = x^2 - 3x$$

Now, putting $p(x) = 0$

$$\Rightarrow x^2 - 3x = 0$$

$$\Rightarrow x(x - 3) = 0$$

$$\therefore x = 0, 3$$

Answer 32: (b) $\frac{1}{\sqrt{3}}$

$$\text{Let, } p(x) = 3x^2 - 1$$

$$\text{Now, } p(x) = 0$$

$$\Rightarrow 3x^2 - 1 = 0$$

$$\Rightarrow 3x^2 = 1$$

$$\Rightarrow x^2 = \frac{1}{3}$$

$$\Rightarrow x = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow x = \frac{1}{\sqrt{3}} \text{ and } -\frac{1}{\sqrt{3}}$$

$\therefore$ (d) is the correct answer.